Two Strategic Sellers without Forgetting

Analytical progress note

October 3, 2026

This note extends the paper's cumulative-learning model by replacing the known outside option with a second long-lived strategic seller. It derives customer demand, state transitions, policy evaluation, best responses, and the equilibrium product-choice conditions. A finite-horizon benchmark is specified analytically; no numerical experiments are undertaken.

Notation. We retain the paper's p_i, c_i, R, γ, its selection, product, and outcome variables y_t, i_t, u_t, and its discounted continuation gap G. Following the previous progress notes, U denotes a value under a fixed policy and V an optimized value. Thus U_j^{σ} will evaluate a fixed policy pair, whereas $V_j^{\sigma_{-j}}$ will denote a best-response value. The pointwise product-choice gain is a_j, extending a_ρ in the previous notes. The symbol π_t^j remains period profit; a deterministic product rule will be denoted by i_j^H in the finite-horizon benchmark. Seller subscripts distinguish the two copies of the original objects.

1 Environment and information

There is one customer and two long-lived strategic sellers, A and B. They have the same products and discount factor:

$$0 < p_1 < p_2 \leq 1, \qquad 0 \leq c_1 < c_2 < R, \qquad \gamma \in (0,1). \tag{1}$$

Product $i \in \{1,2\}$ costs c_i and generates success with actual probability p_i. The selected seller receives revenue R and incurs the cost of its chosen product, regardless of the realized outcome. The unselected seller receives zero and incurs no product cost.

As in the paper, $y_t \in \{A, B\}$ is the seller selected in period t, $i_t \in \{1,2\}$ is that seller's hidden product choice, and $u_t \in \{0,1\}$ is the outcome, with $u_t = 1$ representing success. Periods are indexed by $t = 1, 2, \ldots$. Seller j's period profit is

$$\pi_t^j = \mathbf{1}\{y_t = j\}(R - c_{i_t}), \qquad j \in \{A, B\}, \tag{2}$$

where $\mathbf{1}\{\cdot\}$ is the indicator of an event. There is only one realized product choice i_t in each period.

Both sellers observe the customer's selections and the outcomes generated by either seller. They know the primitives and correctly anticipate each other's equilibrium policy functions. Product choice is private, but the joint observation state is public to both sellers.

The customer observes outcomes and their source, but does not observe the selected seller's product choice. She follows a specified learning and choice rule: separately for each seller j, she treats outcomes as if they were independent Bernoulli observations with a fixed unknown success probability θ_j. This is her statistical representation of that seller's perceived performance or reputation. It is not a claim that the seller's actual success probability is constant under its strategy. She does not infer the sellers' strategies or reconstruct hidden product choices from equilibrium behavior.

2 Timing, state, and customer beliefs

2.1 Timing within a period

The order of events is as follows.

1. The customer independently draws one sample from her current belief about each seller.
2. She selects the seller with the higher sample.
3. Only the selected seller privately chooses product 1 or 2.
4. An outcome is generated with the chosen product's actual success probability:

$$\mathbb{P}(u_t = 1 \mid y_t = j, i_t = i, \text{past observations}) = p_i. \tag{3}$$

5. The selected seller's success or failure count increases by one.

The customer's samples determine selection only. They are not the probabilities used to generate u_t. In particular, sellers do not pay for a product before the customer chooses a seller.

2.2 Cumulative counts and perceived parameters

For seller j, let $S_{j,t}$ and $F_{j,t}$ count the successes and failures observed before period t:

$$S_{j,t} = \sum_{\tau=1}^{t-1} \mathbf{1}\{y_\tau = j, u_\tau = 1\}, \qquad F_{j,t} = \sum_{\tau=1}^{t-1} \mathbf{1}\{y_\tau = j, u_\tau = 0\}. \tag{4}$$

At a generic decision state, suppress the time index and write

$$z = (S_A, F_A, S_B, F_B) \in \mathbb{N}_0^4, \qquad z_0 = (0, 0, 0, 0), \tag{5}$$

where $\mathbb{N}_0$ is the set of nonnegative integers and z_0 denotes the zero-count initial condition. These are ordinary cumulative counts: there is no forgetting, normalization, or reduction to posterior means.

Starting from independent Beta$(1, 1)$ priors, the customer maintains independent belief distributions

$$\theta_A \mid z \sim \text{Beta}(1 + S_A, 1 + F_A), \qquad \theta_B \mid z \sim \text{Beta}(1 + S_B, 1 + F_B). \tag{6}$$

Indeed, in her representation the likelihood for seller j is proportional to $\theta_j^{S_j}(1 - \theta_j)^{F_j}$; multiplying by its uniform prior and normalizing gives (6). Each new outcome updates only the belief associated with its source. The perceived parameters θ_j and the actual product probabilities p_i have different roles and must not be identified.

3 Thompson-sampling demand

Let $\widetilde{\theta}_A$ and $\widetilde{\theta}_B$ be the independent samples from (6). Extending the paper's demand notation,

$$D_A(z) = \mathbb{P}(\widetilde{\theta}_A > \widetilde{\theta}_B \mid z), \qquad D_B(z) = 1 - D_A(z). \tag{7}$$

At every finite state both Beta distributions are continuous, so a tie has probability zero. Both selection probabilities are strictly between zero and one because the belief densities are positive on $(0, 1)$.

For $x \in (0,1)$, define seller j's belief density and cumulative distribution function by

$$f_j(x \mid z) = \frac{x^{S_j}(1-x)^{F_j}}{\mathcal{B}(1+S_j, 1+F_j)}, \qquad H_j(x \mid z) = \int_0^x f_j(b \mid z)\, db. \tag{8}$$

Here $\mathcal{B}(\alpha,\beta) = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\, dx$ is the Beta function, for positive parameters α, β. The letter H for the CDF keeps F_j reserved for failure counts.

Condition first on A's sample being a. Independence makes the conditional chance that B's sample is smaller equal to $H_B(a \mid z)$. Integrating over A's sample gives

$$\begin{aligned} D_A(z) &= \int_0^1 \mathbb{P}(\widetilde{\theta}_B < a \mid z) f_A(a \mid z)\, da \\ &= \int_0^1 f_A(a \mid z) H_B(a \mid z)\, da \\ &= \int_0^1 \int_0^a f_A(a \mid z) f_B(b \mid z)\, db\, da. \end{aligned} \tag{9}$$

Alternatively, conditional on B's sample being b, A wins with probability $1 - H_A(b \mid z)$. Thus the equivalent expression is

$$D_A(z) = \int_0^1 f_B(b \mid z)[1 - H_A(b \mid z)]\, db. \tag{10}$$

Three elementary checks. At the initial state both beliefs are uniform. After one success for A, its density is $2a$; after one failure it is $2(1-a)$. In all three cases B's CDF is $H_B(a) = a$, so

$$\begin{aligned} D_A(0,0,0,0) &= \int_0^1 a\, da = \frac{1}{2}, \\ D_A(1,0,0,0) &= \int_0^1 2a^2\, da = \frac{2}{3}, \\ D_A(0,1,0,0) &= \int_0^1 2(1-a)a\, da = \frac{1}{3}. \end{aligned} \tag{11}$$

These are direct integral calculations, independent of any seller policy.

Comparison with the known outside option. B's posterior mean is $\mathbb{E}[\theta_B \mid z] = (1+S_B)/(2+S_B+F_B)$, where $\mathbb{E}$ denotes expectation. Replacing the paper's known p_0 by this mean would give $1 - H_A(\mathbb{E}[\theta_B \mid z] \mid z)$. Actual demand in (10) is instead the average of $1 - H_A(b \mid z)$ over B's entire belief distribution. These operations are generally different.

For example, at $z = (2,0,0,0)$, A has a Beta$(3,1)$ belief and B a uniform belief. Exactly,

$$D_A(2,0,0,0) = \int_0^1 3a^3\, da = \frac{3}{4}, \qquad 1 - H_A(1/2 \mid z) = 1 - (1/2)^3 = \frac{7}{8}.$$

Thus even this simple state rules out substitution of the posterior mean. If B's belief were instead concentrated at a known p_0, its draw would equal p_0 with probability one, and the comparison would reduce to

$$\mathbb{P}(\widetilde{\theta}_A > p_0 \mid z) = \int_{p_0}^1 f_A(a \mid z)\, da = D(S_A, F_A), \tag{12}$$

the paper's original demand. This recovers the demand comparison; the two-seller game has a different continuation structure.

At a given state, demand is determined by the customer's two belief distributions, not directly by the sellers' current product policies. Product choice takes place after selection and is hidden. Policies affect future demand through the outcomes they generate and the resulting counts.

4 State transitions and calendar time

The superscripts s and f denote success and failure. The four possible state updates are

$$\begin{aligned} T_A^s z &= (S_A + 1, F_A, S_B, F_B), \\ T_A^f z &= (S_A, F_A + 1, S_B, F_B), \\ T_B^s z &= (S_A, F_A, S_B + 1, F_B), \\ T_B^f z &= (S_A, F_A, S_B, F_B + 1). \end{aligned} \tag{13}$$

Define the total number of observations and the individual counts by

$$n(z) = S_A + F_A + S_B + F_B, \qquad n_j(z) = S_j + F_j. \tag{14}$$

For every $j \in \{A, B\}$ and $o \in \{s, f\}$,

$$n(T_j^o z) = n(z) + 1. \tag{15}$$

Starting from z_0, exactly one seller is selected and one outcome is observed each period. Hence n counts completed calendar periods: before period t, $n = t - 1$. In contrast, n_j counts only periods in which seller j was selected. A large $n = n_A + n_B$ need not imply that both individual counts are large.

When B is selected, A's own counts stay fixed, but the joint state changes to $T_B^s z$ or $T_B^f z$. These changes can affect both future demand and both sellers' subsequent product choices. Therefore A's unselected branch cannot be written as $\gamma D_B(z) V_A(z)$. There is no unchanged-joint-state continuation term. The original outside-option denominator $1 - \gamma[1 - D(S, F)]$ came from rearranging such a self-loop and has no corresponding justification here.

5 Policies and fixed-policy evaluation

5.1 Product policies and actual outcome probabilities

A stationary Markov product policy for seller j is a function $\sigma_j : \mathbb{N}_0^4 \to [0, 1]$, with

$$\sigma_j(z) = \mathbb{P}(i_t = 2 \mid y_t = j, z). \tag{16}$$

This probability is the seller's endogenous product policy, conditional on being selected. It is neither demand nor a customer belief. A deterministic policy is the special case $\sigma_j(z) \in \{0, 1\}$ at every state; it need not prescribe the same product at different states.

Under this policy, the actual success probability conditional on selection and the expected product cost are respectively

$$\begin{aligned} q_j(z) &= (1 - \sigma_j(z))p_1 + \sigma_j(z)p_2 = p_1 + \sigma_j(z)(p_2 - p_1), \\ \bar{c}_j(z) &= (1 - \sigma_j(z))c_1 + \sigma_j(z)c_2 = c_1 + \sigma_j(z)(c_2 - c_1). \end{aligned} \tag{17}$$

These are averages over the selected seller's product randomization. Conditional on its realized product i, success still occurs with probability p_i. In particular, $q_j(z)$ is not θ_j, a Thompson sample, or a posterior mean.

Let $\sigma = (\sigma_A, \sigma_B)$ be a fixed pair of policies. The transition probabilities before customer selection are

$$\mathbb{P}(z' = T_k^s z \mid z, \sigma) = D_k(z) q_k(z), \qquad \mathbb{P}(z' = T_k^f z \mid z, \sigma) = D_k(z)[1 - q_k(z)], \tag{18}$$

for $k \in \{A, B\}$, where z' is next period's state. The four probabilities sum to $D_A(z) + D_B(z) = 1$.

5.2 Discounted profits and the evaluation equation

For each $j \in \{A, B\}$, the objective is to maximize

$$\mathbb{E}\left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbf{1}\{y_t = j\}(R - c_{i_t})\right] \tag{19}$$

by choosing its own product strategy, taking the customer's rule and the other seller's strategy as given. Discounting is by calendar period, including periods when the other seller serves the customer.

For a fixed pair σ, define its evaluation value before customer selection by

$$U_j^\sigma(z) = \mathbb{E}_z^\sigma\left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbf{1}\{y_t = j\}(R - c_{i_t})\right]. \tag{20}$$

The subscript z starts the continuation problem at state z; the superscript specifies that both policies are followed thereafter. The expectation averages over customer samples, product randomization, and actual outcomes. Since period profit lies between zero and $R - c_1$,

$$0 \le U_j^\sigma(z) \le \frac{R - c_1}{1 - \gamma}. \tag{21}$$

Separating current profit from next period's value and conditioning on the selected seller gives

$$\boxed{\begin{aligned} U_j^\sigma(z) &= D_j(z)[R - \bar{c}_j(z)] \\ &\quad + \gamma \sum_{k \in \{A,B\}} D_k(z)\left[q_k(z) U_j^\sigma(T_k^s z) + (1 - q_k(z)) U_j^\sigma(T_k^f z)\right]. \end{aligned}} \tag{22}$$

If j is selected, it earns its expected current margin and continues after an update to its own counts. If the other seller is selected, j earns zero now, but its continuation value changes with that seller's outcome. Both branches are discounted by γ because both consume one calendar period. In every continuation term the value belongs to seller j, even when the outcome comes from seller $k \neq j$.

6 Best responses and equilibrium product choice

6.1 Best response to a fixed opposing policy

Hold B's entire function σ_B fixed. Write $V_A^{\sigma_B}(z)$ for A's optimal value before selection when it can choose its product strategy at every future opportunity. In contrast to U_A^σ, this value optimizes A's policy. Formally,

$$V_j^{\sigma_{-j}}(z) = \sup_{\tau_j} \mathbb{E}_z^{\tau_j, \sigma_{-j}}\left[\sum_{t=1}^{\infty} \gamma^{t-1} \mathbf{1}\{y_t = j\}(R - c_{i_t})\right], \tag{23}$$

where $-j$ denotes the other seller and τ_j ranges over admissible product strategies for j, including history-dependent deviations. With the opposing Markov policy fixed, the public state contains all information relevant to rewards and future transitions, so the best-response problem admits the following Markov Bellman formulation.

If A is selected, it compares its two current products using its own optimal continuation value. If B is selected, B uses the fixed $\sigma_B(z)$, with success probability $q_B(z)$. Consequently,

$$\begin{aligned} V_A^{\sigma_B}(z) = D_A(z) \max_{i\in\{1,2\}} \Big\{ R - c_i + \gamma \big[p_i V_A^{\sigma_B}(T_A^s z) + (1-p_i) V_A^{\sigma_B}(T_A^f z) \big] \Big\} \\ + \gamma D_B(z) \big[q_B(z) V_A^{\sigma_B}(T_B^s z) + (1-q_B(z)) V_A^{\sigma_B}(T_B^f z) \big]. \end{aligned} \tag{24}$$

Analogously, holding σ_A fixed gives

$$\begin{aligned} V_B^{\sigma_A}(z) = D_B(z) \max_{i\in\{1,2\}} \Big\{ R - c_i + \gamma \big[p_i V_B^{\sigma_A}(T_B^s z) + (1-p_i) V_B^{\sigma_A}(T_B^f z) \big] \Big\} \\ + \gamma D_A(z) \big[q_A(z) V_B^{\sigma_A}(T_A^s z) + (1-q_A(z)) V_B^{\sigma_A}(T_A^f z) \big]. \end{aligned} \tag{25}$$

The maximum is inside the selected seller's branch because product choice occurs only after selection. Randomizing cannot improve on the larger of the two conditional values; it is optimal only at a tie.

When A deviates, the function σ_B remains fixed, but B applies it to whichever states the deviation generates. For example, $\sigma_B(T_A^s z)$ and $\sigma_B(T_A^f z)$ can differ. Neither B's realized future products nor the future state trajectory is frozen.

For a fixed opposing policy, let $\mathcal{T}_j^{\sigma_{-j}}$ denote the Bellman operator on the right-hand side of the corresponding equation. For any bounded functions v, w, let $\|v - w\|_\infty = \sup_z |v(z) - w(z)|$. Probability weights sum to one, and taking a maximum cannot increase the largest difference between the two product values, so

$$\|\mathcal{T}_j^{\sigma_{-j}} v - \mathcal{T}_j^{\sigma_{-j}} w\|_\infty \le \gamma \|v - w\|_\infty. \tag{26}$$

The bounded-function space is complete, hence this contraction gives a unique bounded best-response value for each fixed opposing policy. The finite product set permits a maximizing stationary selector. The same probability argument gives a unique bounded evaluation value for each fixed pair in (22). These statements do not imply uniqueness of optimal products at ties or uniqueness of the coupled game equilibrium.

6.2 The future value difference and product choice

For a candidate equilibrium pair σ, abbreviate its evaluated values as $V_j = U_j^\sigma$; optimality of these candidate values will be imposed below. As in the paper, define

$$\Delta p = p_2 - p_1 > 0, \qquad \Delta c = c_2 - c_1 > 0. \tag{27}$$

Conditional on seller j being selected, choosing product i now and following the candidate policies thereafter has value

$$Q_j^i(z) = R - c_i + \gamma \left[p_i V_j(T_j^s z) + (1-p_i) V_j(T_j^f z) \right]. \tag{28}$$

The conditional gain $a_j(z)$ from product 2 rather than product 1 is obtained by explicit subtraction:

$$\begin{aligned} a_j(z) &= Q_j^2(z) - Q_j^1(z) \\ &= \left\{R - c_2 + \gamma[p_2 V_j(T_j^s z) + (1-p_2)V_j(T_j^f z)]\right\} \\ &\quad - \left\{R - c_1 + \gamma[p_1 V_j(T_j^s z) + (1-p_1)V_j(T_j^f z)]\right\} \\ &= -(c_2 - c_1) + \gamma\left[(p_2 - p_1)V_j(T_j^s z) + (p_1 - p_2)V_j(T_j^f z)\right] \\ &= -\Delta c + \gamma \Delta p\left[V_j(T_j^s z) - V_j(T_j^f z)\right]. \end{aligned} \tag{29}$$

Product 2 sacrifices Δc in current margin and shifts probability Δp from the failure successor to the success successor. The bracket is the future value difference for seller j. Retaining the paper's discounting convention, its continuation gap is

$$G_j(z) = \gamma\left[V_j(T_j^s z) - V_j(T_j^f z)\right], \qquad a_j(z) = -\Delta c + \Delta p\, G_j(z). \tag{30}$$

Both continuation values allow the two sellers to respond to the new joint state. The formula by itself establishes neither a sign nor a monotonicity property for this future value difference.

To see the cancellation before selection explicitly, define $C_j(z)$ as seller j's discounted continuation contribution when the other seller is selected:

$$C_j(z) = \gamma D_{-j}(z)\left[q_{-j}(z)V_j(T_{-j}^s z) + (1 - q_{-j}(z))V_j(T_{-j}^f z)\right].$$

The two before-selection values for a one-period product deviation differ by

$$\left[D_j(z)Q_j^2(z) + C_j(z)\right] - \left[D_j(z)Q_j^1(z) + C_j(z)\right] = D_j(z)a_j(z). \tag{31}$$

The unselected branch cancels because seller j makes no product choice in that branch. Current demand is identical under the two choices because selection precedes the hidden product choice. Since $D_j(z) > 0$, the conditional and before-selection gains have the same sign. Demand still matters for the levels of all continuation values.

6.3 Stationary Markov perfect equilibrium

A stationary Markov perfect equilibrium is a policy pair $\sigma = (\sigma_A, \sigma_B)$ that is mutually optimal at every state: each seller's policy is a best response to the other's entire policy function, including at states off the equilibrium path. Equivalently,

$$U_A^\sigma(z) = V_A^{\sigma_B}(z), \qquad U_B^\sigma(z) = V_B^{\sigma_A}(z) \qquad \text{for every } z. \tag{32}$$

"Stationary" means the policy depends on the current state rather than separately on calendar time. It does not mean constant products, constant beliefs, or constant market shares. In this cumulative-count model the state also records elapsed time through $n(z)$.

Using the evaluated candidate values in (29), the equilibrium product-choice conditions for each $j \in \{A, B\}$ are

$$\sigma_j(z) = \begin{cases} 1, & a_j(z) > 0, \\ 0, & a_j(z) < 0, \\ \text{any number in } [0,1], & a_j(z) = 0. \end{cases} \tag{33}$$

In the paper's terminology, product 2 is optimal exactly when $G_j(z) \geq \Delta c/\Delta p$, with strict preference when the inequality is strict. This is a product-choice condition, not an assumption that the equilibrium policy has a threshold in a count or a posterior mean.

Conditions (33) must hold together with (22) for *both* sellers at *every* state. They are sufficient as well as necessary: if evaluation and the maximizing product conditions hold, each evaluated value satisfies its own best-response Bellman equation. Uniqueness of that fixed- opponent value then gives (32). At indifference, any mixture is locally maximizing for the given continuation values; changing a mixture in a policy pair still requires consistent reevaluation of both sellers' values.

The sellers have symmetric primitives and initial beliefs, but their realized counts need not be equal. No equality of reputations, constant policy, symmetry restriction on equilibrium, or equilibrium uniqueness is imposed. The separate best-response contractions do not establish uniqueness of a mutually optimal policy pair.

7 An exact finite-horizon benchmark

For a separate finite-horizon game, let the positive integer H be the total number of periods from z_0. Write $V_j^H(z)$ for seller j's equilibrium value before selection with $H - n(z)$ periods remaining, discounted relative to the current period. The horizon H is fixed throughout the recursion. Set

$$V_j^H(z) = 0 \quad \text{when } n(z) = H, \qquad j \in \{A, B\}. \tag{34}$$

Work backwards through layers $n = H-1, H-2, \ldots, 0$. At any state on layer n, all four successors lie on layer $n+1$, so both sellers' continuation values there are already determined.

Conditional on seller j being selected, define

$$Q_j^{H,i}(z) = R - c_i + \gamma \left[p_i V_j^H(T_j^s z) + (1 - p_i) V_j^H(T_j^f z) \right], \qquad i \in \{1, 2\}. \tag{35}$$

For A and B separately, choose the product rule

$$i_j^H(z) \in \arg\max_{i \in \{1,2\}} Q_j^{H,i}(z), \qquad i_j^H(z) = 1 \text{ if } Q_j^{H,1}(z) = Q_j^{H,2}(z). \tag{36}$$

Having chosen *both* products, calculate *both* current values using those same choices:

$$\begin{aligned} V_j^H(z) = {} & D_j(z)[R - c_{i_j^H(z)}] \\ & + \gamma \sum_{k \in \{A,B\}} D_k(z) \left[p_{i_k^H(z)} V_j^H(T_k^s z) + (1 - p_{i_k^H(z)}) V_j^H(T_k^f z) \right], \quad j \in \{A, B\}. \end{aligned} \tag{37}$$

Thus B's selected product affects A's current value through the branch where B serves the customer, and conversely for A's product and B's value. Each seller maximizes its own conditional value, not the other seller's or their sum.

This recursion constructs a pure-strategy Markov equilibrium by backward induction. At the terminal layer there are no further decisions. Given equilibrium continuation play on the next layer, only the selected seller acts at the current state, and its maximizing product is optimal conditional on selection. The other seller has no current decision in that branch. Applying this argument at every state and layer gives sequentially optimal play throughout the finite game. With continuation values fixed, a generic simultaneous 2-by-2 Nash solver is therefore unnecessary. Choosing product 1 at exact indifference makes the recursion definite and selects an equilibrium; it does not establish uniqueness of equilibrium in the game.

In the final decision period, $n(z) = H - 1$, both successor values in (35) vanish. Hence, for each seller,

$$Q_j^{H,2}(z) - Q_j^{H,1}(z) = (R - c_2) - (R - c_1) = -\Delta c < 0. \tag{38}$$

Both sellers therefore prescribe product 1 whenever selected, and $V_j^H(z) = D_j(z)(R - c_1)$ on this layer. This is a terminal-period result for the finite-horizon game. It does not establish eventual extinction of product 2 in the original infinite-horizon model. No convergence of finite-horizon equilibrium policies to a particular infinite-horizon equilibrium is asserted here.

8 What is established and what remains open

The belief rule and timing directly imply the demand integrals, the four state transitions, the fixed-policy and best-response equations, and the product-choice condition in terms of the future value difference. Backward induction gives the stated finite-horizon equilibrium and its final-period use of product 1. These are analytical consequences of the specified model.

Characterizing the infinite-horizon equilibrium policy pair, the shape of its product-choice regions, possible multiplicity, and long-run product use or market shares requires further equilibrium analysis. In particular, the present derivation establishes neither threshold policies nor extinction or persistence of product 2. Numerical analysis, when undertaken separately, may investigate these questions; it would not by itself turn observed patterns into general results.